



EUI BUDDY

Your Smart Campus Assistant!

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INTRODUCTION	• EUIBuddy is a simple yet effective mini chatbot developed in Scala to assist the entire EUI community by providing quick access to essential academic information. It offers details about faculties, campuses, contact information, GPA calculations, and study plans, making it easier for users to find and manage the information they need. 
OBJECTIVES	• Creating an interactive chatbot for the EUI community, providing quick and easy access to academic information related to EUI. • Applying functional programming techniques using Scala , the chatbot is designed with only the necessary features to keep it simple and efficient. • Maintaining a clean, modular, and beginner-friendly design to ensure ease of use for all EUI chatbot users. 
SYSTEM OVERVIEW	• EUIBuddy is a command-driven mini chatbot. • When the program starts, it displays a welcome message and waits for the user to type a natural phrase or instruction asking about anything EUI-related. • The chatbot detects commands by matching keywords against a predefined list stored in a CSV file. • If the command is familiar, it executes the related function and returns a response. • If the command is unfamiliar, it shows an error message and asks the user to retype the question. • Users can also type 'menu' to display a list of all available features the chatbot can perform. • Each feature is implemented as a separate function for better organization. • The chatbot also tracks usage statistics, including the number of commands entered and the top 3 most-used commands, and keeps a history of recent commands to let users revisit previous actions. 

SYSTEM FEATURES	menu - Displays all available features fac - Lists EUI faculties & programs where - Shows campus locations contact - Provides contact details, including hotline, email, & website path - Displays each faculty's study plan & tracks availability grades - Shows the portal link to check results gpa - Calculates term GPA/CGPA based on credit hours & grades stats - Tracks total commands & top 3 searched commands bye - Exits the chatbot & shows a session summary of used commands 
CHALLENGES	• Ensuring proper input validation for user commands and handling invalid inputs effectively. • Managing the chatbot flow and updating responses based on the user's input. • Designing a modular structure so the chatbot remains easy to update and maintain. • Implementing functional programming concepts while keeping the code simple and readable. 
FUTURE DEVELOPMENT	• Adding a graphical interface (GUI) to make the chatbot more user-friendly. • Integrating live data from the official EUI portal for real-time updates. • Expanding the knowledge base to include more academic services and resources. • Implementing multi-language support to assist a wider range of students. • Optimizing chatbot performance for faster responses and better usability. 
CONCLUSION	• The EUIBuddy project shows how a simple Scala 3 chatbot can help users get fast and accurate access to academic information. • By using basic functional programming, the chatbot was built with a clean, organized, and easy to use design. This project creates a strong base for future updates, where more features and better integrations can make EUIBuddy an even more useful tool for all EUI community. 

FP CONCEPTS USED

PURE FUNCTIONS	<i>“Makes the chatbot predictable, testable, and easier to debug.”</i> • def tokenize(input: String): List[String] • def detectCommand(input: String): Option[String] • def getFaculties(): String • def showMenu(): String • def getStats(history: List[(String, String)]): String • def showAnalytics(history: List[(String, String)]): String
HIGHER-ORDER FUNCTIONS	<i>“Simplifies the code and makes calcs like GPA more concise and functional.”</i> <pre>(1 to numCourses).foldLeft((0.0, 0.0)) { case ((pointsAcc, creditsAcc), i) => val credit = readDoubleSafe(s"Course \$i credit hours: ") val grade = readGradeSafe(s"Course \$i grade: ") (pointsAcc + credit * grade, creditsAcc + credit)}</pre>
IMMUTABILITY	<i>“Ensures data consistency and prevents further modifications.”</i> • val faculties = Map("CIS" -> "...", "ENG" -> "...", ...) • val campuses = Map("Knowledge City" -> "New Administrative Capital") • val contacts = Map("Hotline" -> "10071-0001", ...)
PATTERN MATCHING	<i>“Improves code readability and simplifies command input handling.”</i> <pre>val response = finalCommand match { case "greet" => ... case "menu" => showMenu() case "fac" => getFaculties() case "gpa" => calculateGPA() case _ => "Sorry, I don't understand."}</pre>
RECURSION	<i>“Replaces traditional loops with a cleaner and more functional approach.”</i> <pre>def askFaculty(): String = { val input = StdIn.readLine().trim.toUpperCase input match { case "CIS" "ENG" "BI" "DA" => input case _ => println("Invalid Faculty."); askFaculty() } }</pre>

Note: Some functions in EUIBuddy are impure because they perform I/O operations or loading data from CSV files. Functions such as `startChat()`, `calculateGPA()`, and `askFaculty()` require side effects, so they cannot be purely functional.

TEST CASES

FACULTIES

Welcome to EUIBuddy - Your Smart campus Assistant!
Type freely, and I'll understand your request.

You: HEY
EUIBuddy:
HI! Need any assistance?

You: Yes can I know what majors in EUI?
EUIBuddy:
CIS -> Computing & Information Systems
ENG -> Engineering
BI -> Business Informatics
DA -> Digital Arts & Design
(Double-check the official EUI pages; details may change.)



LOCATION

You: Where is the campus location ?
EUIBuddy:
Knowledge City -> New Administrative Capital
NAC -> New Administrative Capital
(Double-check the official EUI pages; details may change.)



CONTACT

You: How can I contact EUI?
EUIBuddy:
Hotline: 10071-0001
Email: info@eui.edu.eg
Website: <https://eui.edu.eg>
(Double-check the official EUI pages; details may change.)



GRADES

You: How to check my marks ?
EUIBuddy:
Check your results on the EUI student portal: <https://eui.edu.eg/>
(Double-check the official EUI pages; details may change.)



GPA CALCULATOR

```
You: Can you calculate my GPA ?  
How many courses did you study this term? 3  
Course 1 credit hours: 4  
Course 1 grade (A+, A, B, ... , F): A  
Course 2 credit hours: 3  
Course 2 grade (A+, A, B, ... , F): B-  
Course 3 credit hours: 3  
Course 3 grade (A+, A, B, ... , F): C+  
Your term GPA is: 3.10  
What was your previous CGPA? 2.7  
How many total credit hours have you completed? 55  
EUIBuddy:  
Your term GPA is: 3.10  
Your updated CGPA is: 2.76
```



STUDY PLAN

```
You: Can you tell me more about EUI's study plan ?  
Which faculty's study plan would you like to see? (CIS, ENG, BI, DA)  
CIS  
EUIBuddy:  
  
Computing & Information Sciences Study Plan:  
  
Duration: 4 Years (8 Terms)  
Structure:  
- Divided into 4 academic years, each with 2 terms.  
- Courses are taken in sequence, with prerequisites for advanced classes.  
- Meeting with an academic advisor helps tailor the plan to your goals.  
  
Available Tracks:  
1. Computer Science  
2. Artificial Intelligence  
3. Cyber Security  
4. Data Science & Engineering  
5. Software Engineering (coming soon)  
6. Graphics & Game Programming (coming soon)  
  
(Double-check the official EUI pages; details may change.)
```



USAGE STATISTICS

```
You: Can you give me a session summary ?  
EUIBuddy:  
Total commands: 7  
Top commands:  
Can you calculate my gpa ? -> 1 times  
How to check my marks ? -> 1 times  
What about the campus location ? -> 1 times
```



EXIT CHATBOT

```
You: BYE  
GOODBYE!  
  
Session Summary:  
Total commands used: 9  
Most used commands:  
Can you calculate my gpa ? -> 1 times  
How to check my marks ? -> 1 times  
What about the campus location ? -> 1 times
```

